

Viral Infections and the Impact on Oral Health

Integrated Biomedical and Clinical Sciences 2: OMFS 513

Rashidah Wiley, DDS



Mission: To advance the science and art of health through education, service, scholarship and social accountability
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Learning Outcomes & Competency Statements

Learning Outcomes

1. Recognition of the different types of viruses.
2. Identify the oral and systemic manifestation associated with viral infections.

Competency statements met

1.3, 2.1, 3.1, 4.1 & 5.1

Human Herpes Viruses

Double-stranded DNA, enveloped viruses from the Herptoviridae family.

Modes of transmission: saliva, respiratory droplets, and sexual contact.

1. Human Simplex Virus-1 (HSV-1)
2. Human Simplex Virus-2 (HSV-2)
3. Human Herpes Virus-3 (Varicella-Zoster Virus-VZV)
4. Human Herpes Virus-4 (Epstein-Barr Virus-EBV)
5. Human Herpes Virus-5 (Cytomegalovirus-CMV)
6. Human Herpes Virus-6
7. Human Herpes Virus-7
8. Human Herpes Virus-8 (Kaposi sarcoma-associated herpes virus-KSVH)



Acute Herpetic Gingivostomatitis

- Human Herpes Virus 1
- More common in children and young adults
 - Activation of the virus after initial exposure

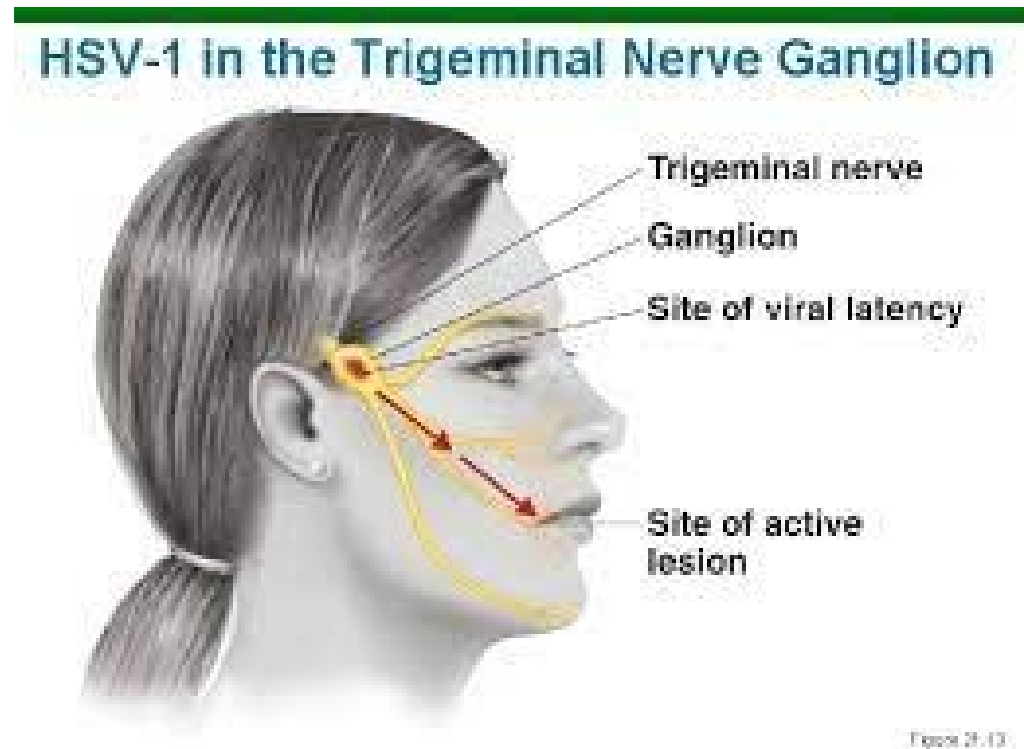
1. Primary Infection

2. Latency

- the virus lies dormant in the sensory nerve until activated
- Most common site of latency in H&N: **trigeminal ganglion**
- Once activated, patients experience extensive oral lesions and flu-like symptoms

3. Recurrence

- After the 1st episode the virus lies dormant and becomes reactivated during times of immunosuppression or stress



Acute Herpetic Gingivostomatitis

- **Symptoms**

- Fever, headache, irritability
- Lymphadenopathy
- Acute gingivitis- ulcers of the interdental papilla and attached gingiva
- Crops of vesicles
 - The vesicles burst leaving behind painful red erosions

- **No dental treatment with active lesions**

- Herpetic whitlow

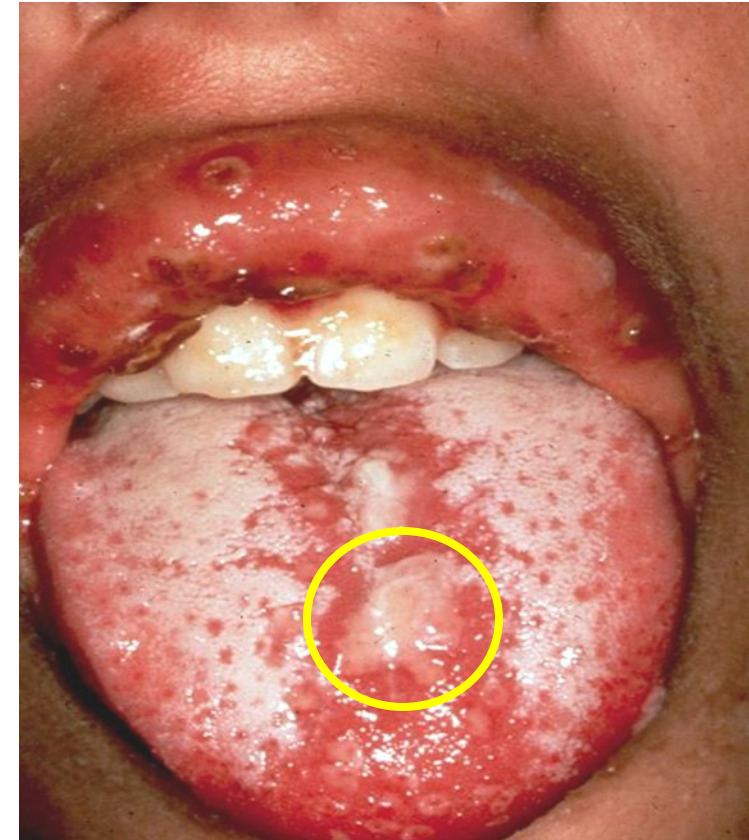
- **Treatment**

- Antiviral therapy- Acyclovir
- Palliative care:
 - Rest and take in fluids. Supportive medications to help with symptoms



www.orthobullets.com/hand/6100/herpetic-whitlow

Acute Herpetic Gingivostomatitis



Courtesy of Oral Pathology Lab Inc.

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Recurrent Herpetic Infection (Secondary Herpes)

- Due to a reactivation of HSV-1 or 2
- Occur at the site of inoculation or area of mucosa near the involved nerve
- Prodromal symptoms:
 - Burning and tingling
- Common locations on **keratinized mucosa**
 - Vermilion border and adjacent skin
 - Attached (bound down) oral mucosa
 - Genitalia

Recurrent Herpetic Infection

- **Clinical appearance**
 - Appear as crops of vesicles
 - Heals within 7-10 days spontaneously
- **Treatment**
 - Antivirals
 - Palliative care

Recurrent Herpetic Infection



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Varicella-Zoster Virus (HSV-3)

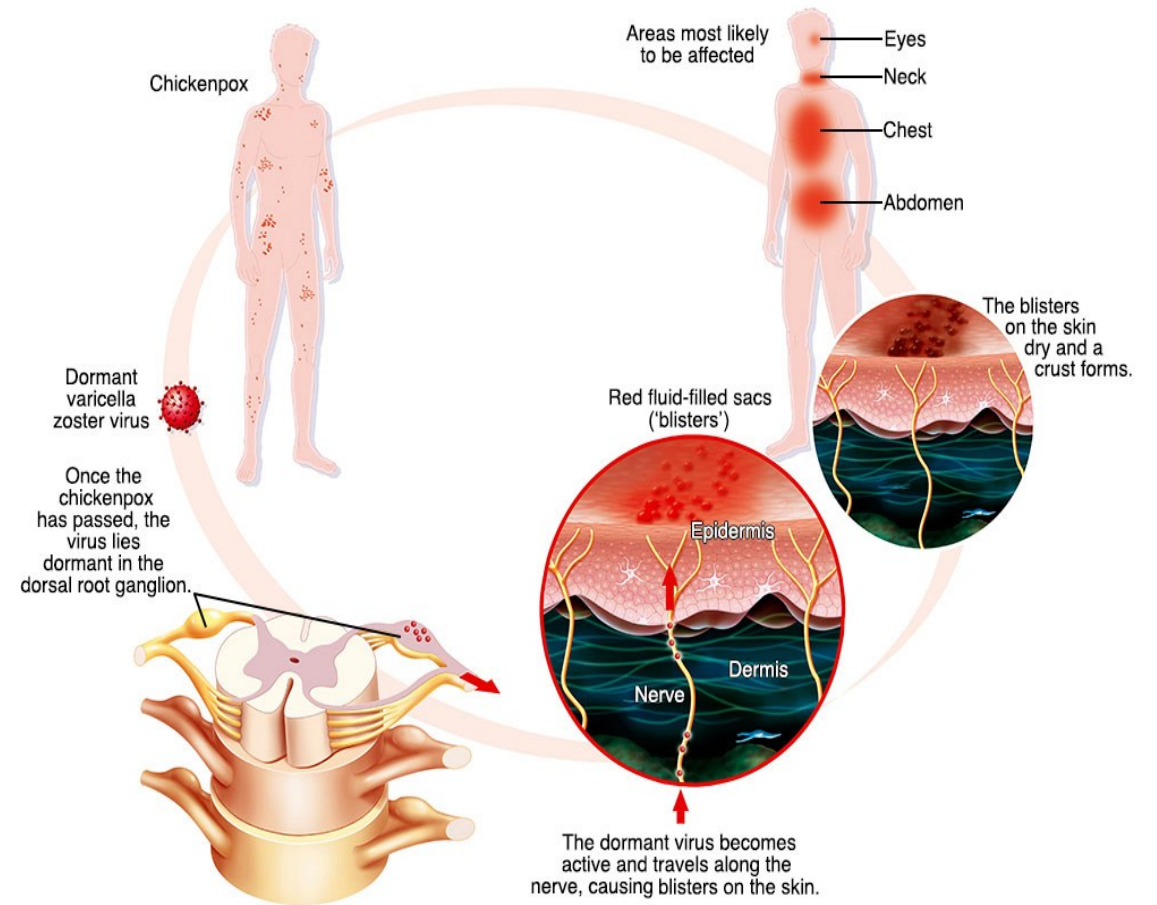
- Transmission: Droplets and direct contact
- Chicken Pox and Shingles
- **Symptoms**
 - Malaise, pharyngitis, and rhinitis
- Skin rash
 - Vesicles that rupture and crust over
- Oral
 - Small ulceration
- Older and immunocompromised patients may experience complications with chicken pox

Chicken Pox on the Skin



Zoster/Shingles (HHV-3)

- Occurs years after the initial infection
 - Virus is dormant in the dorsal root ganglion before reactivation
- Involves the affected sensory nerves (dermatomes)
- Pain for 3-5 days over the affected area
- Unilateral linear vesicular eruptions
- Ocular involvement



Unilateral eruption of shingles on the scalp and Face



Unilateral shingles of the labial mucosa



Oral and Maxillofacial Pathology, 3rd Edition

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Zoster/Shingles (HHV-3)

- **Ramsay Hunt Syndrome**
 - Ipsilateral facial paralysis
 - Shingles of the face and external auditory canal
 - Loss of taste to the anterior 2/3rd of the tongue
- **Chronic form-Post herpetic neuralgia**
 - Pain lingers 3 months after the acute rash
 - Affects adults aged 60 and older
- **Treatment**
 - Prevention: ProQuad, Varivax, VariZIG, Shingrix, and Zostavax
 - Antiviral therapy
 - Palliative Care

BACKGROUND

- * RARE COMPLICATION of VARICELLA-ZOSTER VIRUS REACTIVATION
 - ~ OCCURS in LESS THAN 1% of INFECTIONS
 - ~ CONGENITAL or ACQUIRED LATER in LIFE
 - ~ MORE OFTEN in ELDERLY & IMMUNOCOMPROMISED

CAUSES

- * VARICELLA-ZOSTER VIRUS (VZV)
 - ~ CAUSES SHINGLES & CHICKENPOX
 - ~ CAN REMAIN DORMANT in CRANIAL NERVES & DORSAL ROOT GANGLIA
 - ~ STRESSFUL CONDITION or ILLNESS may CAUSE REACTIVATION & REPLICATION, WHICH TRAVELS through NERVE FIBERS ASSOCIATED with INVOLVED GANGLION

SIGNS & SYMPTOMS

- * FACIAL NERVE PALSYP
 - ~ WEAKNESS on ONE SIDE of FACE
 - ~ DIFFICULTY SMILING, FROWNING, RAISING EYEBROWS, PUFFING CHEEKS, or WHISTLING
 - ~ LOSS of NASOLABIAL FOLD or WRINKLING in FOREHEAD COMPARED to UNAFFECTED SIDE
 - ~ DIFFICULTY CLOSING EYES, ESPECIALLY during SLEEP (LEADS to ↑↑ RISK of KERATOPATHY)
- * EAR PAIN
- * VESICULAR RASH in AURICLE or EAR CANAL
 - ~ FLUID-FILLED VESICLES MAY APPEAR on TONGUE & PALATE
- * AFFECTS GENICULATE GANGLION CAUSING SYMPTOMS along FACIAL NERVE
- * MAY OCCUR with RASH KNOWN as "ZOSTER SINE HERPETE"
- * OTHER CRANIAL NERVES MAY ALSO BE AFFECTED
 - ~ VESTIBULOCOCHLEAR NERVE: HEARING LOSS, TINNITUS, VERTIGO
 - ~ VAGUS NERVE: HOARSENESS OR ASPIRATION

TREATMENT

- * ↓↓ DURATION of SYMPTOMS & COMPLICATIONS
- * MEDICATIONS
 - ~ ANTIVIRALS
 - ~ CORTICOSTEROIDS
 - ~ MOST EFFECTIVE within 72 HOURS of SYMPTOM ONSET
- * VACCINATION against VARICELLA-ZOSTER VIRUS
- * ARTIFICIAL TEARS during DAY & LUBRICATING OINTMENTS at NIGHT to PROTECT CORNEA
- * SEVERE FACIAL PARALYSIS:

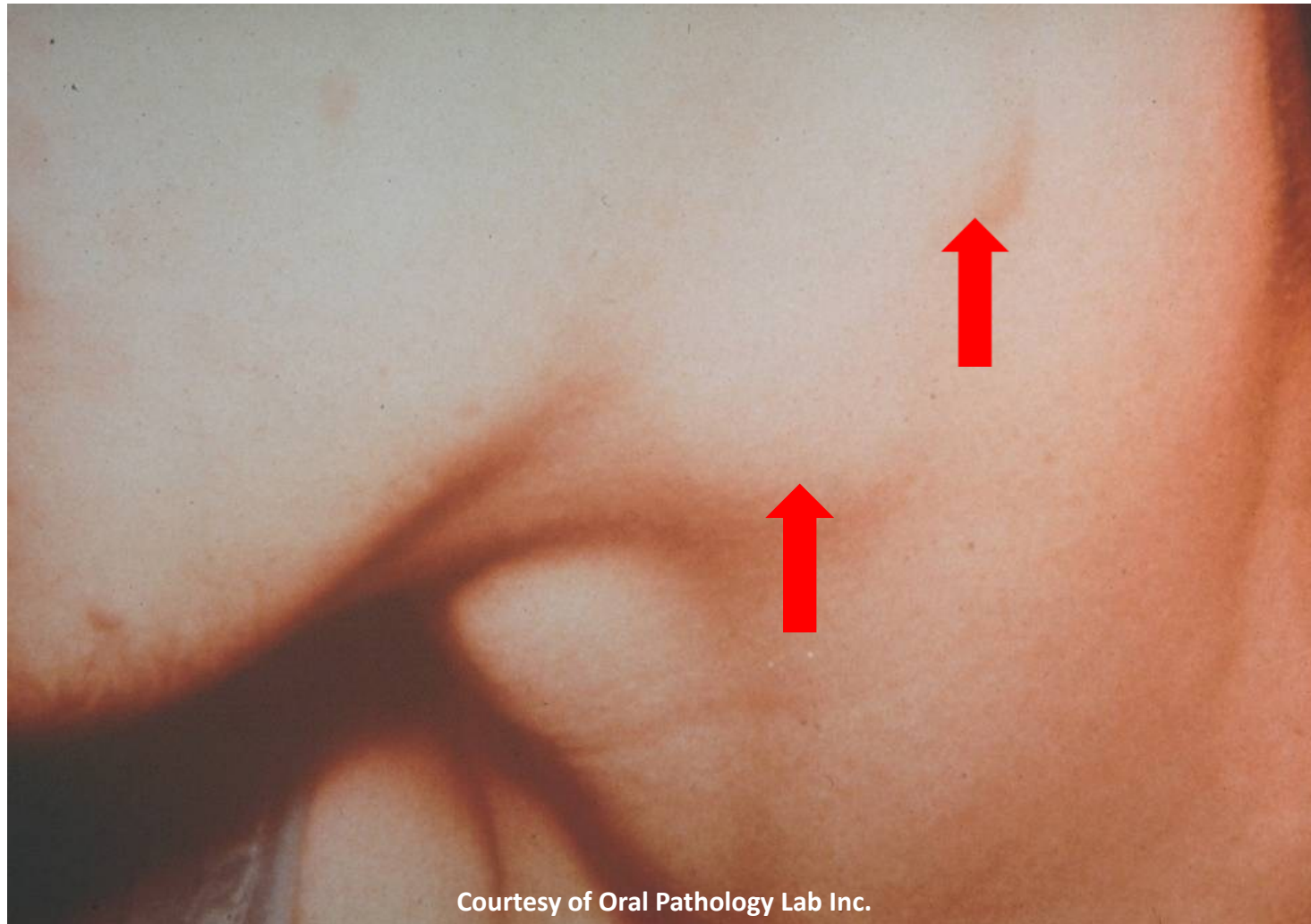
Epstein-Barr Virus (HHV-4)

- Diseases associated with EBV
 - **Infectious mononucleosis**
 - Oral hairy leukoplakia
 - Nasopharyngeal carcinoma
 - Specific types of Hodgkin's and Non-Hodgkin's Lymphoma
 - **Burkitt's Lymphoma**

Infectious Mononucleosis “Kissing Disease”

- Epstein Barr Virus (HSV-4/EBV)
- Mode of transmission: direct saliva transfer
- Increased white blood count (WBC), normal platelet count
- Symptoms:
 - Pharyngitis, lymphadenopathy, fever, cough with petechiae on the soft palate
- Management:
 - Rest, fluid intake, analgesics, and antipyretics

Cervical Lymphadenopathy with Mono



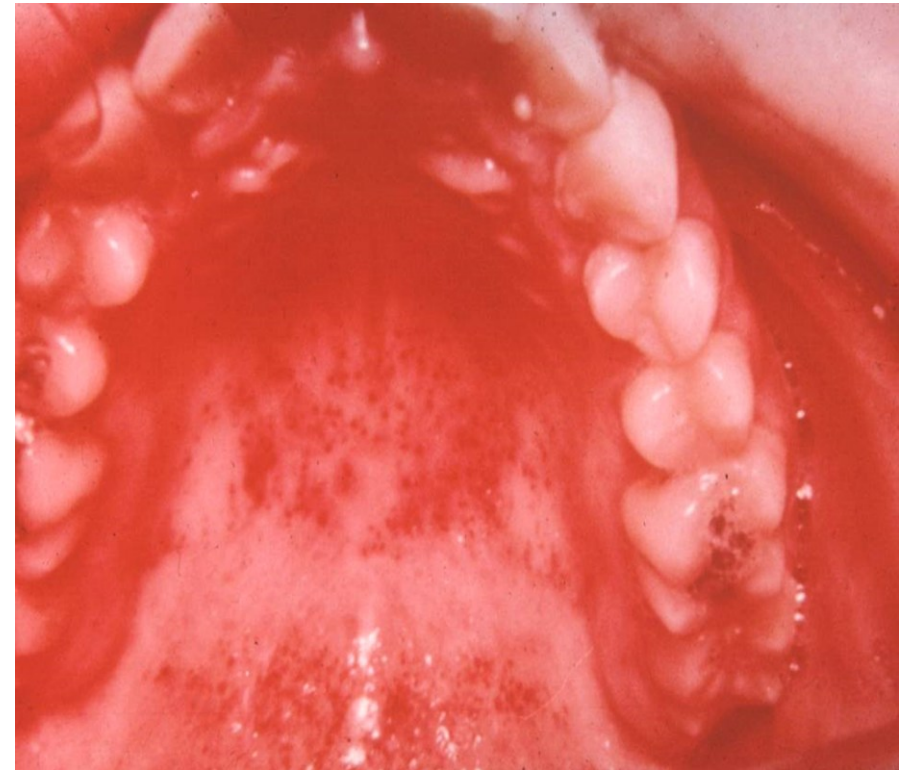
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Infectious Mononucleosis



Bilateral tonsillar enlargement with erythema and exudate on the soft tissues



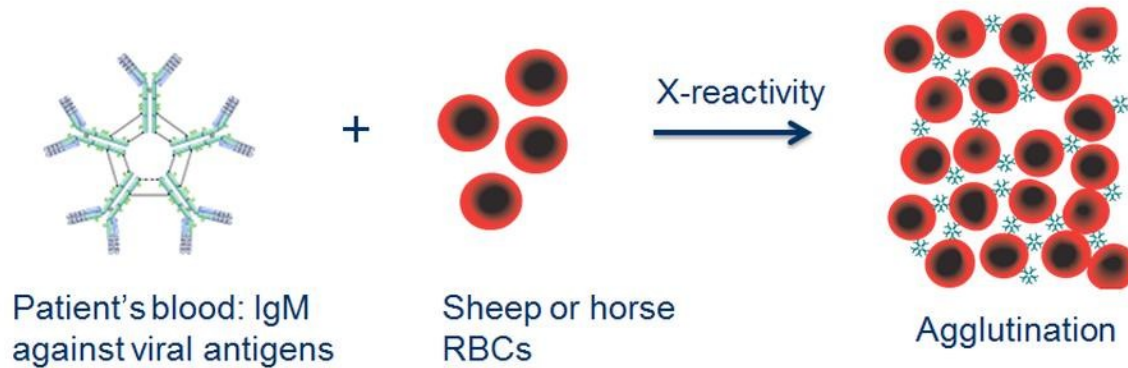
Petechiae noted on the hard and soft palate

Courtesy of Oral Pathology Lab Inc.

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Diagnostic Test for Mono: Monospot Test

Heterophile Antibody (Monospot)

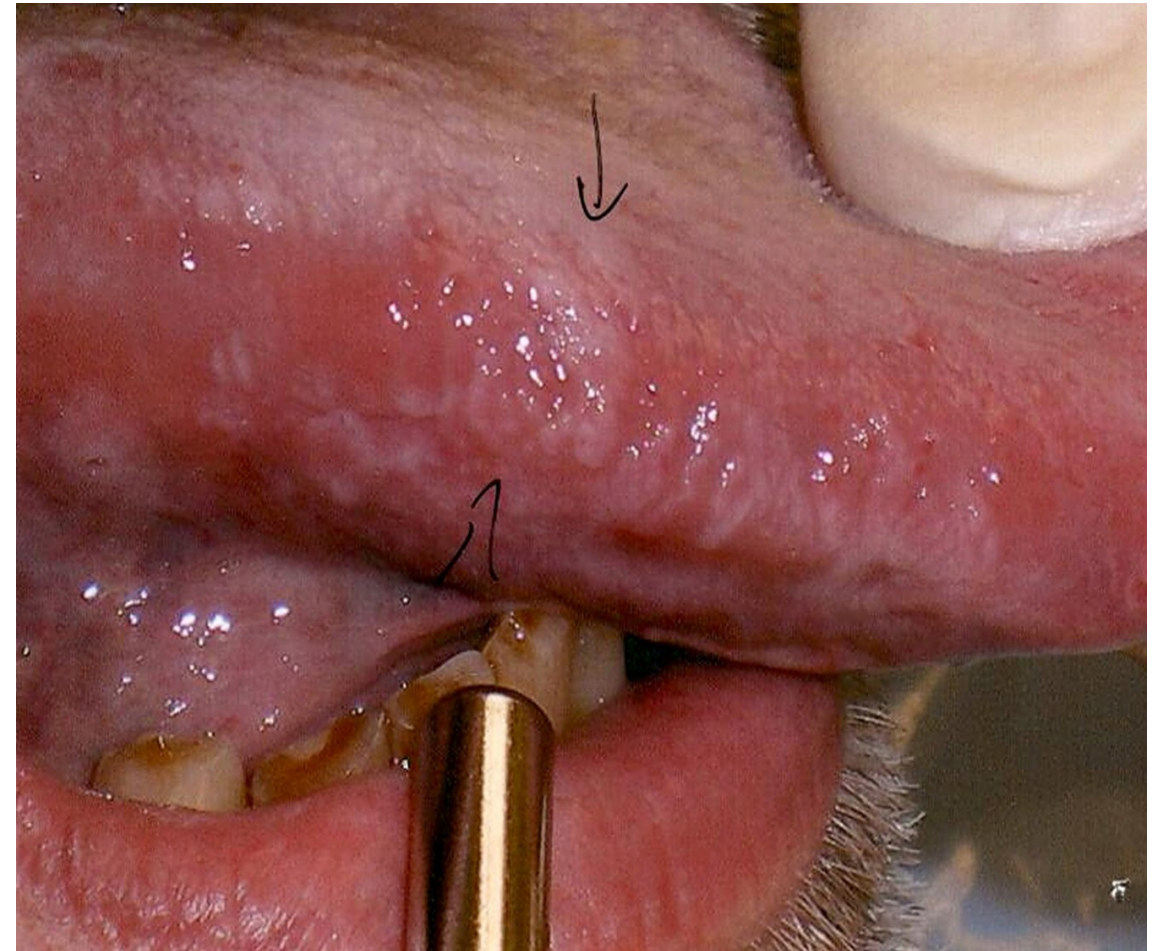


- Sensitivity: 90-95% after 1st week, but only 75% in the 1st week
- Specificity: 94%. False (+): infections (CMV, Toxo, **acute HIV**), malignancy, autoimmune (SLE)
- *No longer recommended by the CDC given false (+) and (-)*

- Current practices use the patient's blood to test for antibodies against the EBV itself or the capsid surrounding the virus.
- A CBC with differential lab test will also be obtained.
- Other clinical test will be completed to rule out differential diagnoses.

Oral Hairy Leukoplakia (HHV-4)

- Association with HIV+ patients and other immunocompromised individuals.
- A small subset set of cases can be seen in immunocompetent patients.
- Clinical presentation
 - Vertical white plaques on the lateral border of the tongue



Courtesy of Oral Pathology Lab Inc.

Cytomegalovirus (CMV)

- HHV-5
- **Mode of transmission**
 - Close contact, bodily fluids, and birth
- **Clinical presentation**
 - Infectious mononucleosis
 - Ocular involvement: loss of vision
 - Oral: chronic oral ulcerations and enamel hypoplasia
 - Salivary gland enlargement
- **Treatment:** Antiretroviral therapy

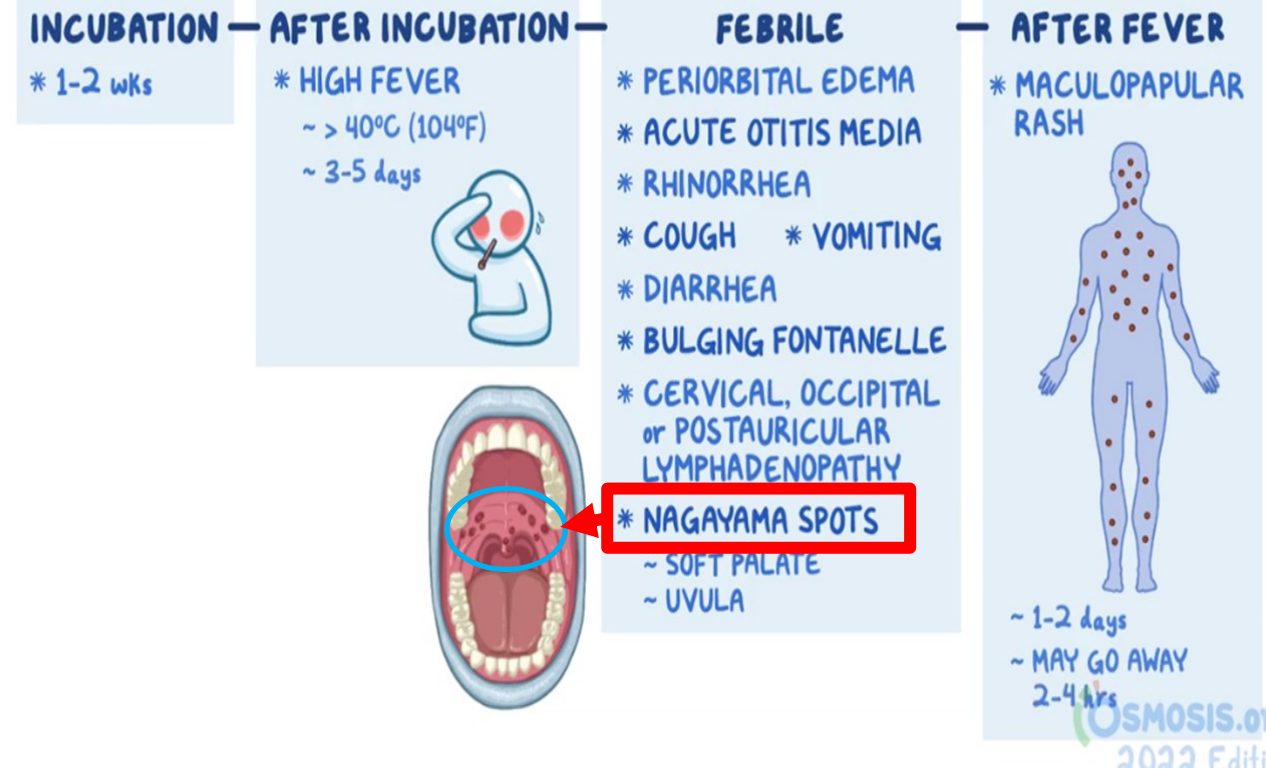


Courtesy of Oral Pathology Lab Inc.

Human Herpes Viruses 6 and 7

- Roseola/sixth disease
- Spread through salivary and respiratory droplets
- Virus establishes latency in CD4+ cells
- Reactivation of the virus can affect to multiorgan infection.

ROSEOLA INFANTUM



Roseola Infantum

TREATMENT

* SUPPORTIVE

~ ROSEOLA → SELF-LIMITING DISEASE



FEVER



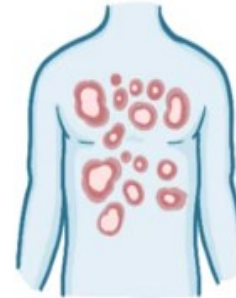
ACETAMINOPHEN
or
IBUPROFEN



VOMITING or
DIARRHEA



FLUID
INTAKE



RASH



GOES AWAY
w/o TREATMENT



IMMUNOSUPPRESSION



ANTIVIRALS
~ ACYCLOVIR
~ GANCICLOVIR

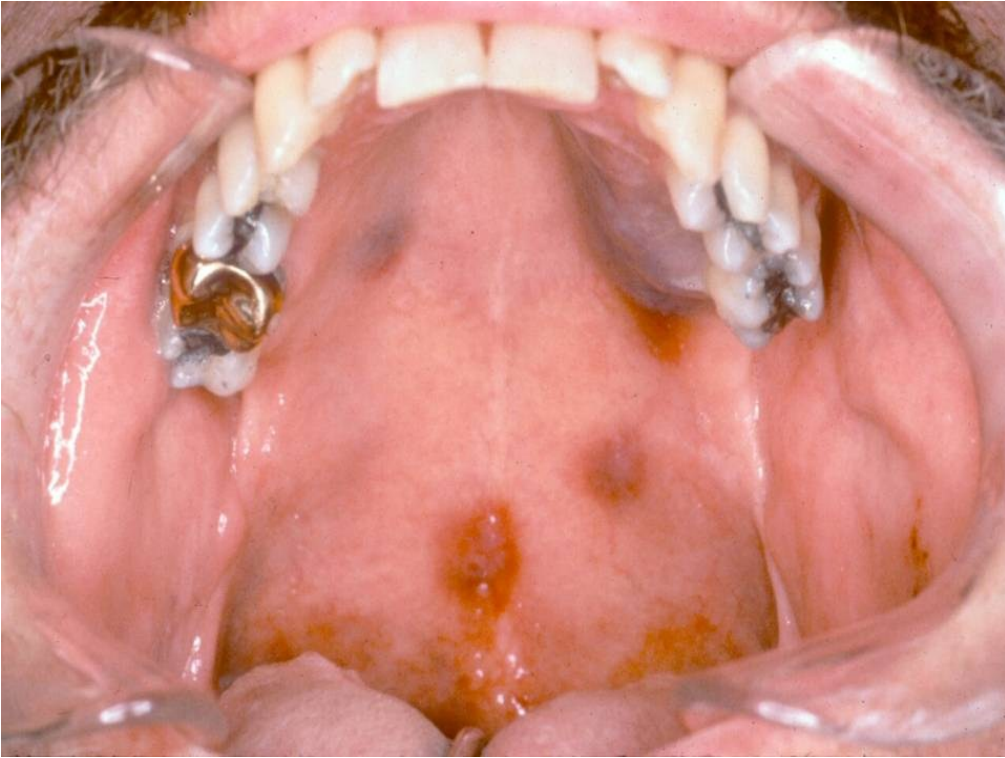


~~Aspirin in children~~

Kaposi Sarcoma (HHV-8)

- 4 Clinical Variants
- Male predilection
- Skin: Purple macules or nodules
- Oral: Dark reddish-brown masses

Kaposi Sarcoma



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Rubeola (9-Day Measles)

- Morbillivirus
- Transmission via respiratory droplets
- 3 stages:
 1. Incubation stage
 2. Prodromal stage: fever, conjunctivitis, rhinitis, and coughing
 3. Exanthem: Rash develops in head and neck region before spreading to the rest of the body
- Oral lesions: Koplik Spots
- Other symptoms: Diarrhea and pneumonia
- MMR-V vaccine

Koplik Spots on the Buccal Mucosa



Jayasree P, Ashique KT, Kaliyadan F. Koplik Spots. JAMA Dermatol. 2021 Apr 1;157(4):456

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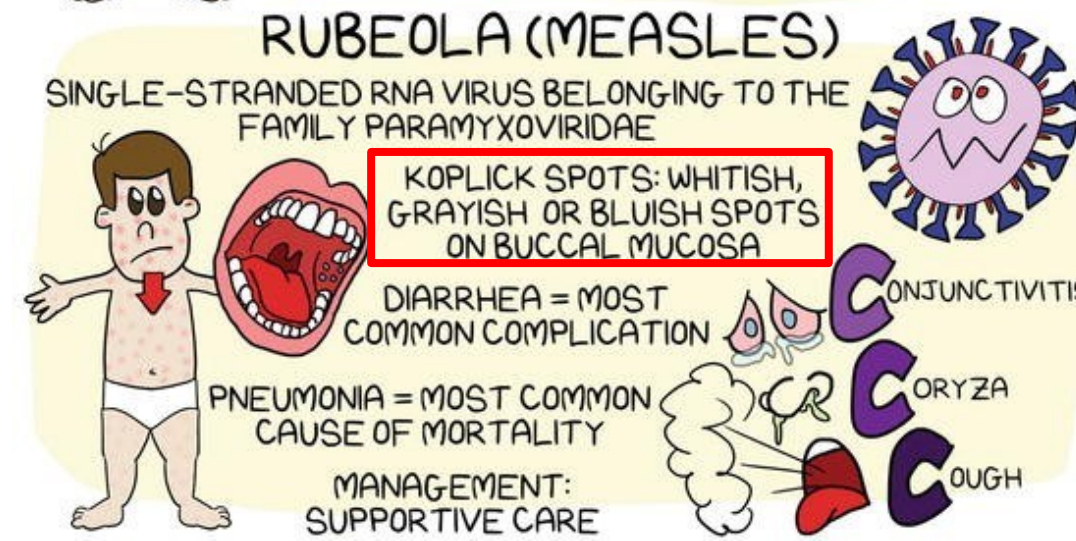
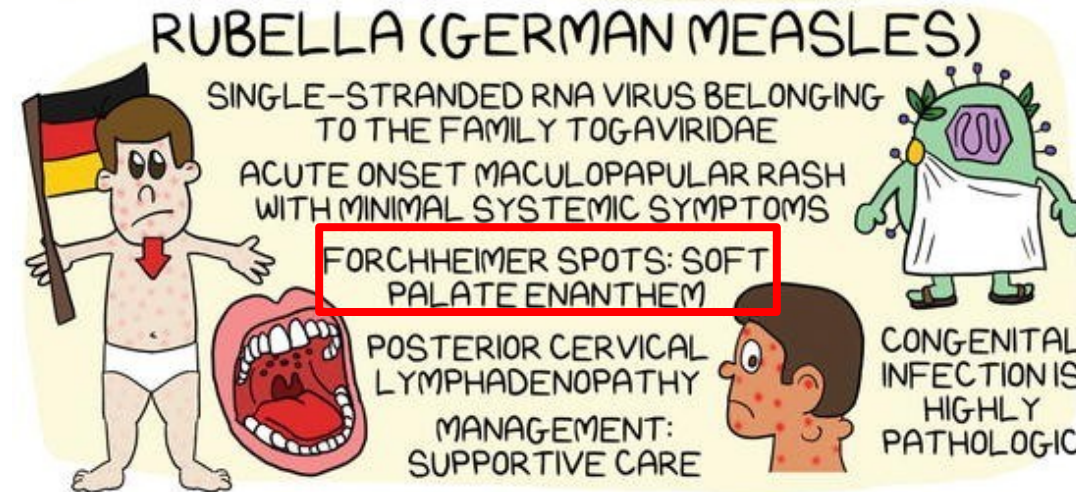
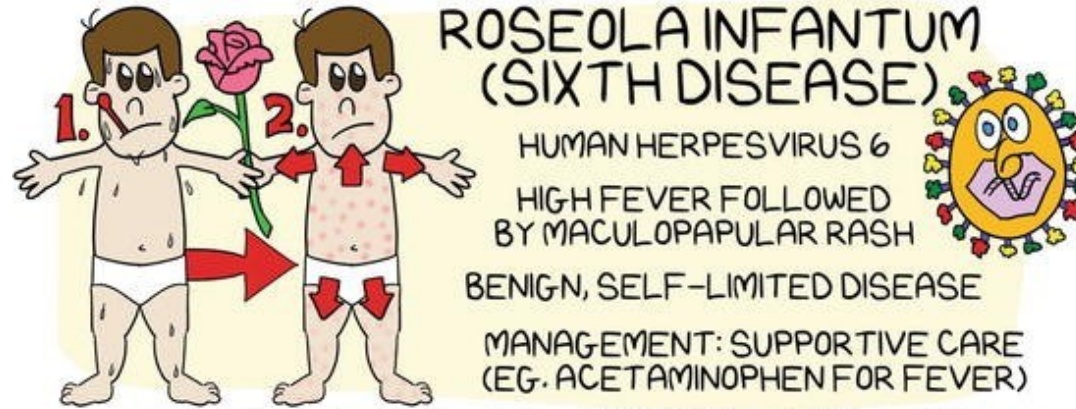
Rubella (German Measles)

- Togavirus
- Transmission via respiratory droplets
- Exanthematous rash on the face and neck
 - **Resolution in 3 days**
- Arthritis is a common complication

Rubella (German Measles)

- Oral lesions
 - **Forchheimer's sign**
 - Dark red papules found on the soft palate
- Congenital Rubella Syndrome
 - Deafness (80%)
 - Heart Disease
 - Cataracts





Enteroviruses

- Non-enveloped, RNA viruses
 - Coxsackieviruses, echoviruses, and enteroviruses
 - Mode of Transmission: Fecal-Oral Transmission
 - GI- issues
 - Rash of skin and/or oral mucosa
 - Flu-like symptoms
1. Herpangina
 - Acute Lymphonodular Pharyngitis
 2. Hand-Foot-Mouth Disease



Herpangina

- Flu-like, sore throat, and GI symptoms
- Acute lymphonodular pharyngitis
 - Variant of Herpangina
- Small number of red lesions seen on the tonsillar pillars 1-2 mm in size
 - Heal with in 1-2 weeks



Consultant 360 Vol. 2 (6) June 2003.



Mohammed Malik Afroz et al. Int. J. Res. Pharm. Sci., 2021. 12(1). 392-395.

Hand-Foot-Mouth Disease (HFMD)

- Most well-known of the enterovirus
- Skin rash on palms and feet
- Flu-like and GI symptoms
- Oral lesions
 - Red macules which form vesicles then ulcerate
 - Sites: buccal mucosa, labial mucosa and tongue
 - Last for 1 week

Hand-Foot-Mouth Disease (HFMD)



<http://www.vnews.mv>



www.news4jax.com/health

Enteroviruses

	Herpangina	Herpes gingivostomatitis	Hand foot and mouth disease	Bacterial pharyngitis	Aphthous stomatitis	Infectious Mononucleosis
Causative organism	Enteroviruses	Herpes Simplex Virus 1 and 2	Enteroviruses	Group A streptococcus	-/+*	Ebstein Barr Virus
Oral lesions	+	+	+	+	+	+/-
Vesicular/Ulcerative lesions	+	+	+	-	+/-	+/-
Anterior pharynx	-	+	+	-	+	+/-
Posterior pharynx	+	+/-	-	-	-	+/-
Tonsillar exudates	-	-	-	+	-	+
Cervical lymphadenopathy	+/-	+/-	+/-	+	-	+
Fever	+	+	+	+	-	+
Systemic symptoms	-	+	-	+	+*	+

*depends on the etiology

Corsino CB, Ali R, Linklater DR. Herpangina. In: StatPearls. StatPearls Publishing, Treasure Island (FL); 2021.

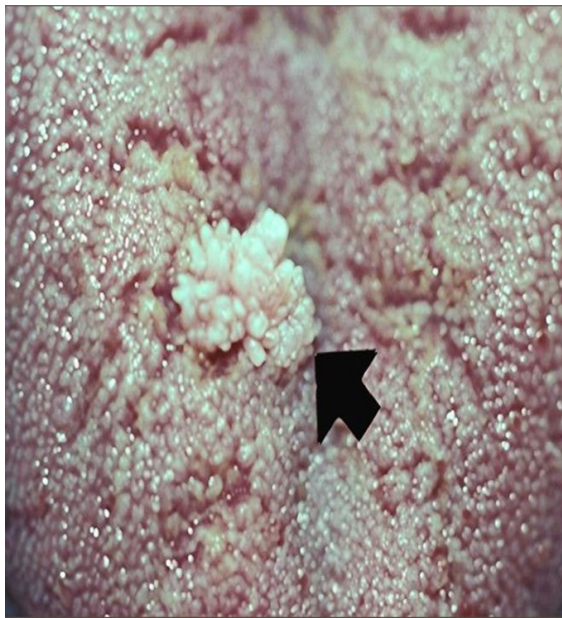
Epidermic Parotitis (Mumps)

- Rubulavirus
- Transmission via urine, saliva, and respiratory droplets
- Clinical symptoms
 - Flu-like symptoms
 - Swelling of the salivary glands and testis
 - Complication in male - sterility



Human Papilloma Virus (HPV)

- Non-enveloped, double-stranded DNA Virus
 - Associated with warty lesions of the mucosa and skin
 - Squamous Papilloma
 - Verruca Vulgaris
 - Multifocal Epithelial Hyperplasia (Heck's Disease)
 - Condyloma acuminatum
 - Malignant transformation can be seen with certain viruses in this family
 - **HPV 16**
- 



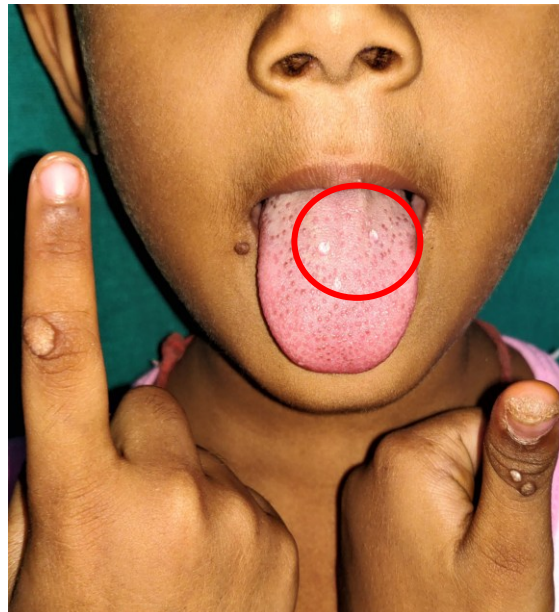
Squamous Papilloma



Condyloma Acuminatum



Multifocal Epithelial Hyperplasia



Verruca Vulgaris "common wart"

Table 1

Classification of HPV Types Based on Cervical Cancer Risk

High risk	16, 18, 31, 33, 35, 39, 45, 51, 52, 56, 58, 59, 68, 73, 82
Probable high risk	25, 53, 56
Low risk	6, 11, 40, 42, 43, 44, 54, 61, 70, 72, 81

Source: Reference 25.

Resources

- Osmosis.org
- Ebell, Mark H. "Epstein-Barr virus infectious mononucleosis." American family physician vol. 70,7 (2004): 1279-87.

